WEEKLY LITERATURE REVIEW REPORT

Project: Post-Disaster Assessment using ML and Semantic Communication

Timeline: Week 2

1. KEY PAPER SUMMARY

- Paper Title: Robust Disaster Assessment from Aerial Imagery Using Text-to-Image Synthetic Data
- Core Methodology Studied: The authors proposed a scalable pipeline leveraging text-guided mask-based image editing capabilities of generative diffusion models. They created thousands of post-disaster images from low-resource domains synthetically and implemented a simple two-stage training approach combining manual supervision from source domains with synthetic target data.
- Relevance to Our Project: This paper provides an exceptional solution to our project's core challenge regarding class imbalances and data scarcity (such as our limited samples for damaged roads and garbage pollution). It demonstrates that text-to-image generative models can synthetically enrich minority classes while preserving domain robustness. This serves as a vital literature foundation for our plan to solve the sparse data problem before training our classification models.

2. IN-TEXT CITATION EXAMPLE

"To circumvent the severe scarcity of manually labeled data in localized or under-resourced geographies, text-guided generative models can be deployed to synthetically scale up post-disaster datasets and improve domain robustness."

REFERENCES (IEEE Format)

T. Kalluri, J. Lee, K. Sohn, S. Singla, J. Xu, J. Liu, and M. Chandraker, "Robust Disaster Assessment from Aerial Imagery Using Text-to-Image Synthetic Data," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2024, pp. 1-10.

2. KEY PAPER SUMMARY

• Paper Title: Attention Based Semantic Segmentation on UAV Dataset for Natural Disaster Damage Assessment

• Core Methodology Studied: The authors implemented a novel self-attention based semantic segmentation neural network architecture tested on a high-resolution Unmanned Aerial Vehicle (UAV) dataset to identify multiple damaged structures like buildings and roads. Their self-attention mechanism captures global contextual dependencies, achieving a Mean IoU score of approximately 88%.

• Relevance to Our Project: Since our fieldwork heavily utilizes high-resolution localized aerial data from drone imagery, this study justifies the transition from satellite observations to tactical UAVs. Furthermore, understanding how self-attention weights capture broad visual patterns helps us map out what semantic features are most vital to extract at our UAV edge device before transmitting classification tags over limited network channels.

2. IN-TEXT CITATION EXAMPLE

"Implementing self-attention schemes on high-resolution Unmanned Aerial Vehicle (UAV) data allows neural networks to map out global contextual structures, significantly enhancing pixel-level damage segmentation boundaries."

REFERENCES (IEEE Format)

[2] T. Chowdhury and M. Rahnemoonfar, "Attention Based Semantic Segmentation on UAV Dataset for Natural Disaster Damage Assessment," in arXiv preprint arXiv:2105.14540v2, 2021, pp. 1-5.

3. KEY PAPER SUMMARY

• Paper Title: Cross-directional Feature Fusion Network for Building Damage Assessment from Satellite Imagery

• Core Methodology Studied: The authors introduced a cross-directional feature fusion strategy designed to exploit deep correlations between pre-disaster and post-disaster imagery. Additionally, they integrated the CutMix data augmentation technique to specifically combat the challenge of hard-to-classify or imbalanced structural damage targets.

• Relevance to Our Project: While this study focuses on large-scale xBD satellite imagery, its technical insights regarding the CutMix augmentation strategy are highly relevant to our project. It gives us a validated mathematical blueprint for tackling hard classes and minority feature sets within our Roboflow workspace, enabling more stable multi-class classification mappings.

2. IN-TEXT CITATION EXAMPLE

"By leveraging cross-directional feature fusion networks along with CutMix data augmentation, deep learning pipelines can better resolve localized structural variances across pre- and post-disaster scene matrices [3]."

REFERENCES (IEEE Format)

Y. Shen, S. Zhu, T. Yang, and C. Chen, "Cross-directional Feature Fusion Network for Building Damage Assessment from Satellite Imagery," in arXiv preprint arXiv:2010.14014v2, 2020, pp. 1-8.